

Elbee Racing — Data Acquisition Feature Report

Generated 2026-08-06 00:45 from KinematiK · Electrics subsystem.

Data Acquisition — results

Result	Value	Status
Worst-case load	35.9%	
Messages	40	

Data Acquisition — checks & verdicts

- ✗ BLOCKED — 2 hard failure(s) Something in this plan does not work.
- • Motor stator temperature mounts on the motor, which is on the tractive-system side of the isolation boundary. Its signal path into the grounded low-voltage system needs a galvanic barrier. This is not an EMC preference — an un-isolated sensor wire is a conductive path across the boundary, and it will be found at scrutineering if it is not found earlier.
- • Inverter / IGBT temperature mounts on the inverter, which is on the tractive-system side of the isolation boundary. Its signal path into the grounded low-voltage system needs a galvanic barrier. This is not an EMC preference — an un-isolated sensor wire is a conductive path across the boundary, and it will be found at scrutineering if it is not found earlier.
- ✗ Damper position, front left: sampling at 50 Hz a signal with content to 30 Hz (1.7x). Everything above 25 Hz folds down and appears as a lower frequency that is indistinguishable from real data. This is not noise you can filter out afterwards — the information is destroyed at the ADC. Sample at 150 Hz or above.
- ⚠ Damper position, front left is an analog channel with no declared anti-alias filter. The filter is what makes the sample rate meaningful; without one, any interference above half the sample rate (switching noise, inverter carrier, RF) folds into the band you care about and cannot be separated from it afterwards.
- ⚠ Wheel speed FL: 2.5x oversampling clears the theorem but not practice. Peaks land between samples and the trace shape is unreliable. 800 Hz would let you read this channel off a plot with confidence.
- ⚠ Wheel speed FR: 2.5x oversampling clears the theorem but not practice. Peaks land between samples and the trace shape is unreliable. 800 Hz would let you read this channel off a plot with confidence.
- ⚠ Wheel speed RL: 2.5x oversampling clears the theorem but not practice. Peaks land between samples and the trace shape is unreliable. 800 Hz would let you read this channel off a plot with confidence.
- ⚠ Wheel speed RR: 2.5x oversampling clears the theorem but not practice. Peaks land between samples and the trace shape is unreliable. 800 Hz would let you read this channel off a plot with confidence.
- • Brake pressure, front circuit: 2 of 9 applicable review questions unanswered — What connector / wiring does it need?; How do we test / calibrate it?.
- ⚠ Brake pressure, front circuit is marked as estimated — the numbers are placeholders, not datasheet values, and every budget below inherits that.
- ⚠ Brake pressure, front circuit has no part number or datasheet link. A specification with no source cannot be checked by the next person, and the next person is you in October.

- **⚠** Brake pressure, front circuit has no owner. Unowned channels are the ones that arrive unmounted.
- **✗** Tractive system current mounts on the accumulator, which is on the tractive-system side of the isolation boundary. Its signal path into the grounded low-voltage system needs a galvanic barrier. This is not an EMC preference — an un-isolated sensor wire is a conductive path across the boundary, and it will be found at scrutineering if it is not found earlier.
- **⚠** GNSS position: 2.0x oversampling clears the theorem but not practice. Peaks land between samples and the trace shape is unreliable. 50 Hz would let you read this channel off a plot with confidence.
- **✗** The BMS frame (204 bytes at 10 bits/byte) needs 212% of 9600 baud to sustain 10 Hz. The link physically cannot carry it. Either raise the baud rate or accept a lower update rate — and note that a BMS often will not let you do the first.
- **✗** 1 signal(s) are scheduled onto CAN faster than the BMS produces them (10 Hz): pack_current. The extra frames would repeat stale values, which looks like fresh data on a plot and is not.
- **⚠** Shutdown-relevant signals (cell_v_min, cell_v_max, cell_t_max) reach the bus 213 ms after the BMS measures them. That is fine for logging and too slow to be part of any protective action — the shutdown circuit must remain hardware, not a CAN message.
- **✗** 'coolant_temp_in' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.
- **✗** 'wheel_speed_fl' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.
- **✗** 'wheel_speed_fr' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.
- **✗** 'wheel_speed_rl' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.
- **✗** 'wheel_speed_rr' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.
- **✗** 'brake_pressure_f' cannot reliably complete transmission within its own period at this bus load. Its worst-case arbitration delay exceeds the interval before the next sample is due, so samples will be overwritten in the transmit buffer before they reach the bus.

Data Acquisition — charts & tables

Message	ID	DLC	Rate (Hz)	Bits	Load	Producer
motor_temp	0x400	2	10	75	0.15%	R. Okafor
coolant_temp_in	0x401	2	200	75	3.00%	P. Lindqvist
coolant_temp_out	0x402	2	5	75	0.07%	P. Lindqvist
coolant_flow	0x403	2	20	75	0.30%	P. Lindqvist
damper_pot_fl	0x404	2	50	75	0.75%	M. Haddad
steer_angle	0x405	2	100	75	1.50%	M. Haddad
wheel_speed_fl	0x406	2	200	75	3.00%	M. Haddad

wheel_speed_fr	0x407	2	200	75	3.00%	M. Haddad
wheel_speed_rl	0x408	2	200	75	3.00%	S. Varga
wheel_speed_rr	0x409	2	200	75	3.00%	S. Varga
brake_pressure_f	0x40a	2	200	75	3.00%	daq
ts_current	0x40b	2	500	75	7.50%	A. Ferreira
BMS_50Hz_0	0x300	2	50	75	0.75%	bms_bridge
BMS_10Hz_1	0x301	6	10	115	0.23%	bms_bridge
BMS_10Hz_2	0x302	8	10	135	0.27%	bms_bridge
BMS_10Hz_3	0x303	8	10	135	0.27%	bms_bridge
BMS_10Hz_4	0x304	8	10	135	0.27%	bms_bridge
BMS_10Hz_5	0x305	8	10	135	0.27%	bms_bridge
BMS_10Hz_6	0x306	8	10	135	0.27%	bms_bridge
BMS_10Hz_7	0x307	8	10	135	0.27%	bms_bridge
BMS_10Hz_8	0x308	8	10	135	0.27%	bms_bridge
BMS_10Hz_9	0x309	8	10	135	0.27%	bms_bridge
BMS_10Hz_10	0x30a	8	10	135	0.27%	bms_bridge
BMS_10Hz_11	0x30b	8	10	135	0.27%	bms_bridge
BMS_10Hz_12	0x30c	8	10	135	0.27%	bms_bridge
BMS_10Hz_13	0x30d	8	10	135	0.27%	bms_bridge
BMS_10Hz_14	0x30e	8	10	135	0.27%	bms_bridge
BMS_10Hz_15	0x30f	8	10	135	0.27%	bms_bridge
BMS_10Hz_16	0x310	8	10	135	0.27%	bms_bridge
BMS_10Hz_17	0x311	8	10	135	0.27%	bms_bridge
BMS_10Hz_18	0x312	8	10	135	0.27%	bms_bridge
BMS_10Hz_19	0x313	8	10	135	0.27%	bms_bridge
BMS_10Hz_20	0x314	8	10	135	0.27%	bms_bridge
BMS_10Hz_21	0x315	8	10	135	0.27%	bms_bridge
BMS_10Hz_22	0x316	8	10	135	0.27%	bms_bridge
BMS_10Hz_23	0x317	8	10	135	0.27%	bms_bridge
BMS_10Hz_24	0x318	8	10	135	0.27%	bms_bridge
BMS_10Hz_25	0x319	8	10	135	0.27%	bms_bridge
BMS_10Hz_26	0x31a	2	10	75	0.15%	bms_bridge
BMS_1Hz_27	0x31b	1	1	65	0.01%	bms_bridge

40 rows